ADITYA UNIVERSITY

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PART-I

S.No: BT1030

Programe	:	B.Tech
Hall Ticket No:		25B11EC816
Name	:	VENNAPUSA MOHAN REDDY
Examination	:	B.Tech I Semester - END EXAMINATIONS Regular
Month-Year	:	Dec-25
Branch	:	Electronics and Communication Engineering
Course Code	:	2501CH02
Course Name	:	APPLIED CHEMISTRY
Date of Exam	:	05-01-2026

J. Pavan.
Signature of the Controller of Examination

V. Mohan Reddy 5/1/26.
Signature of the Student with date

Sindhu 5/1/26
Signature of the Invigilator with date

Serial No. Last Page Written	15

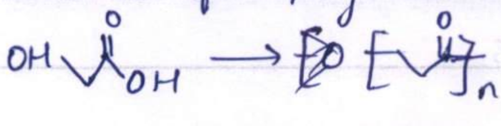
INSTRUCTIONS TO STUDENTS

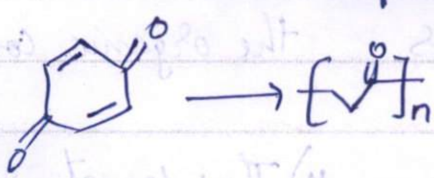
1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

Q.No.

2) a) which are degradable in soil using bacteria and microorganisms is called degradable.
Eg: Bakelite, PGA etc

Process:
1) It obtained from polymerisation of glycolic acid


2) It can also be obtained from glycolidic acid


Properties:
1) PGA is naturally occurring polymer in the nature
2) PGA is adsorbable
3) It has boiling point of 225-230°c
4) It can be breakdown into glycolic acid using enzymes and etc
5) It breakdowns into glycolic acid which is naturally formed in our body
6) It has tential stress and modulus of elasticity

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2)b)

Thermoplastic	Thermo setting
1) It is obtained by addition polymerisation	1) It is obtained by condensation polymerisation
2) It doesnot have cross linked structures and only chain structures	2) It is a three dimensional structure
3) It is soluble in organic compounds	3) It is insoluble in most of the organic compounds
4) They can be reused, re shaped and recycled	4) They cannot reused and reshaped
5) It is soft, weak and less brittle	5) It is strong, hard and brittle
Eg:- Bakelite	Eg:- Poly Vinylchloride (PVC)

1)a) Polymer:- polymer is defined as the it is the structural chain of monomers which forms polymers

Properites:-

1) It is very cheap, easy to preapare
2) It can synthesis easily

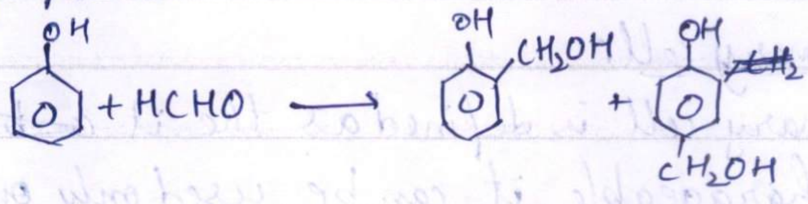
Q.No.

3) They can easily transported
4) They are very light in weight and durable.
5) These can be moulded in any shape

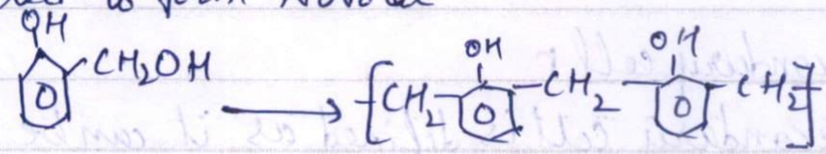
1)b) Bakalite: It is first synthesised polymer invented in 1907 by leo Baekelite.

Bakelite is prepared in different steps

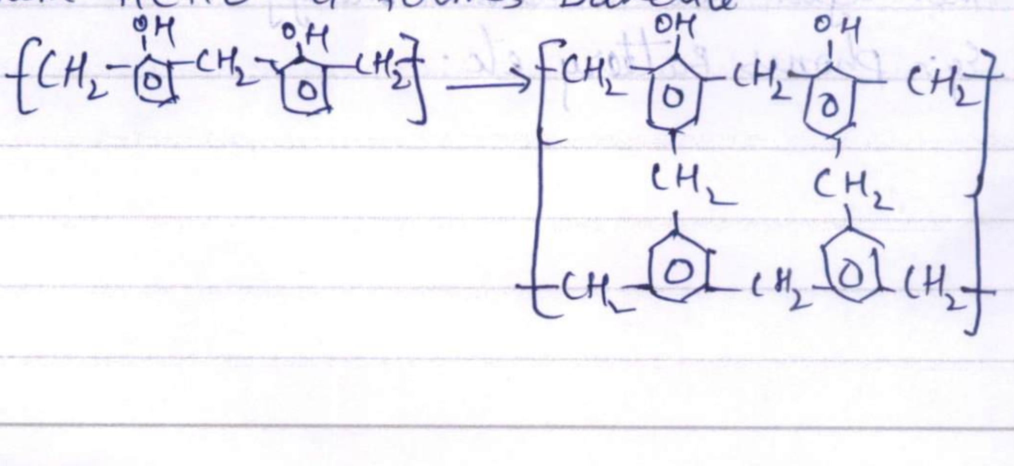
step 1: phenol is reacted with formaldehyde in presence of acid / alkaline medium to form o- and p- hyroxyl methyl phenol



step 2: These o- and p- hydroxyl methyl phenol is heated to form Novolac



step 3: Novolac on further heating and by reacting with HCHO it forms Bakelite



Properties :-

1) It is rigid, hard, scratch resistant, infusable, water resistemt, fire resistant
2) It has a good adhesive nature
3) It ~~is~~ a good insulator of electricity

Applications :-

1) It is used to make switches, boards, electronics frames etc.
2) In Varnishes and paints
3) It is used in moulds to makes T.V, phones parts

3) a) Primary cell :-

Primary cell is defined as the it ~~can be~~ cann't be recharageable it can be used only once.
Eg:- Watch Batteries, wall clock Batteriees etc.

Secondary cell :-

Secondary cell is defined as it can be recharageable it converts mechanical energy into electrical energy and then again into mechanical energy.
Eg:- phones Battery etc.

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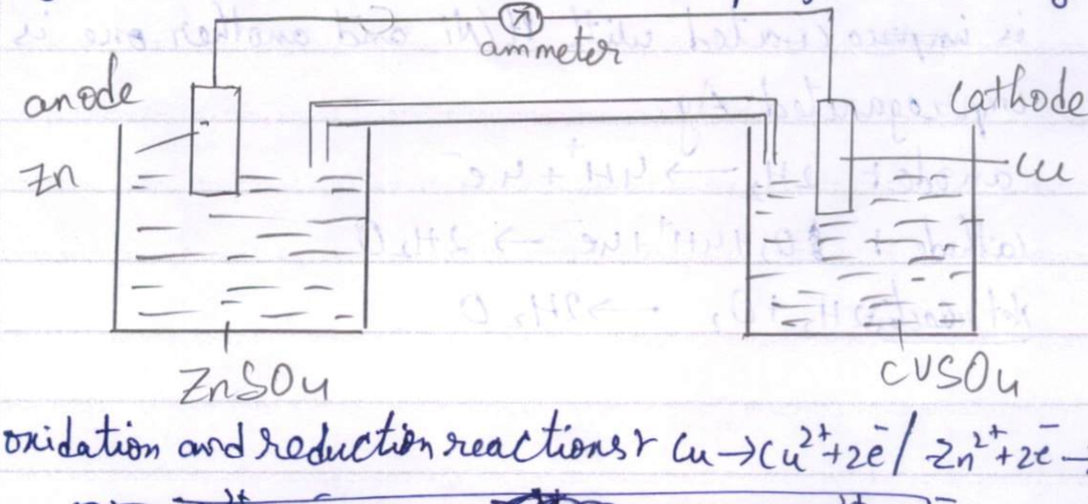
Primary cell	Secondary cell
1) It cannot be rechargeable	1) It can be rechargeable
2) It has only limited power and one time use	2) It also have limited power but can be charged many times
3) It cannot convert energy from one form to another	3) It can convert energy from one form to another
Eg: torch, wallclock batteries	Eg: Phones, powerBank batteries

electro chemical cell:
It is defined as it converts energy from chemical energy in to Electrical energy

It can be formed by using Zn rod in $ZnSO_4$ solution and Cu plate in $CuSO_4$ solution where Zn acts as anode and Cu acts as cathode. rods are connected by external wire and internally by salt bridge.



oxidation and reduction reactions: $Cu \rightarrow Cu^{2+} + 2\bar{e}$ / $Zn^{2+} + 2\bar{e} \rightarrow Zn$

~~$Cu^{2+} + 2\bar{e} \rightarrow Cu$~~ ~~$Zn \rightarrow Zn^{2+} + 2\bar{e}$~~

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Here Zn ions from anode moves to cathode to form Zn^{2+} ions and sulphate ions in cathode moves to anode by this exchange of ions forms the electricity in the circuit.

Principle:

By the above function states that "An operation that happens to Exhange of ions in circuit forms electricity this is called voltaic cell(or) chemical cell(or) glavanic cell"

3)b) ~~H_2~~ fuel-cell:

In fuel cell fuel is converted into electrical energy. fuel ~~is taken~~ is H_2-O_2 (H_2-O_2 are taken

H_2-O_2 fuel cell:

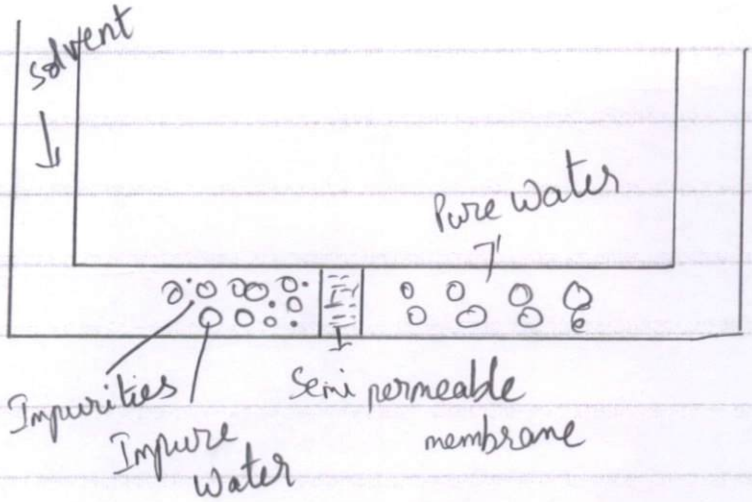
Here we take two porous carbon electrode one acts as anode and one acts as cathode Here H_2 & O_2 are libreated at their respective electrode

one electrolyte is use 5M KOH is used. The cathode is ~~impreg~~ coated with Pt/Ni and another one is impreganted Ag.

anode: $2H_2 \longrightarrow 4H^+ + 4e^-$

cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

Net reaction: $2H_2 + O_2 \longrightarrow 2H_2O$


solvent
Pure water
Impurities
Impure water
Semi permeable membrane

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